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Solar PV-diesel hybrid systems for the Nigerian private sector: An impact assessment

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ABSTRACT

This research examines the impact of Nigerian private sector investment in captive power generation from solar photovoltaic (PV) and diesel generator (DG) hybrid energy systems. The study assesses the economic viability of solar PV-DG hybrid systems among Nigerian private companies using levelized cost of energy (LCOE) and analyzes policies that can facilitate solar PV investment as a bottom-up approach to Nigeria's energy development. Forty (40) private companies across Nigeria were surveyed to examine potential the fuel savings that hybrid of PV and DG can achieve. Out of the surveyed companies, three cases representing different load factors were selected and modelled in HOMER Pro to obtain LCOE for each case study. The research findings indicate that electricity generation deficiencies for Nigerian business consumers can be addressed with available renewable energy solar resources. A savings between € 0.002 and 0.009/kWh was achieved from the three-case study. Adequate policy to support investment is both necessary and possible for achieving increased solar PV adoption for highly willing Nigerian private sector.

1. Introduction

With a growing population of over 180 million people and availability of many natural resources, Nigeria has great potential for development. The country's vision is to be among the leading 20 economies across the globe by the year 2020 (INTEC, 2015). Nigeria was once expected to be among the next global economic powerhouses alongside Mexico, Indonesia and Turkey, with all being grouped as “MINT” (Elliot, 2014). However, this potential has remained inert. As 75% of its budget is from oil revenue, Nigeria is currently struggling to survive with oil prices below US$100 (World Energy Council, 2016). At present, Nigeria is still far from finding its niches in global economy in the presence of its multi-faceted problems and Nigeria running on a mono-economy is not sustainable.

Energy availability is key and is currently lacking in Nigeria (see Fig. 1), and the power sector remains a key constraint to achieving economic development (INTEC, 2015). Nigerian possess enormous energy resources from both renewable and non-renewable sources. The main source of power in the country is from fossil fuel (INTEC, 2015), which is highly susceptible to the risk of fuel price volatility, availability and other socio-political forces that can create insecurity (Ata, 2015). The total electricity generation in Nigeria has a share of about 85% from natural gas (Edomah, 2019) while the rest is from hydropower resources (Edomah, 2019; ICREEE, 2016; PWC, 2016). Notwithstanding, frequent gas shortage for thermal power plants and occurrence of drought against hydropower operations stand as a big impediment to optimal utility of these electric infrastructures. An empirical case of that is high generation slide from 1085 kW to 172 kW (Asu, 2016) by Egbin thermal power plant in 2016, which was supplies 32% of Nigerian electricity (Aliyu et al., 2013). For Nigeria to ensure multi-dimensional growth, infrastructure development such as power plants generation is considered one of five key areas that must be addressed (Gravito et al., 2016) to provide reliable and affordable power supply (Ikeme and Ebohon, 2005).

Frequent grid power outages leading to grid defection and immense diesel generator usage (Edomah, 2019; Emodi and Boo, 2015), rising costs of fuel and increasing costs of operating and maintaining electric generators has caused hardship for many private business enterprises (Akwagyiram and George, 2016). Between 2000 and 2014, no fewer than 2000 factories were forced to shut down due to energy challenges (Anyagafu, 2014), indicating the gravity of this problem. This infrastructural deficiency is prevalent in many countries in Sub-Sahara Africa, posing difficulties to businesses. Solar PV may help industries in Nigeria as well as other African countries to achieve substantial fuel savings from the use of diesel generators (DG), with up to 30% of annual diesel consumption depending on insolation level (Bopp, 2015).

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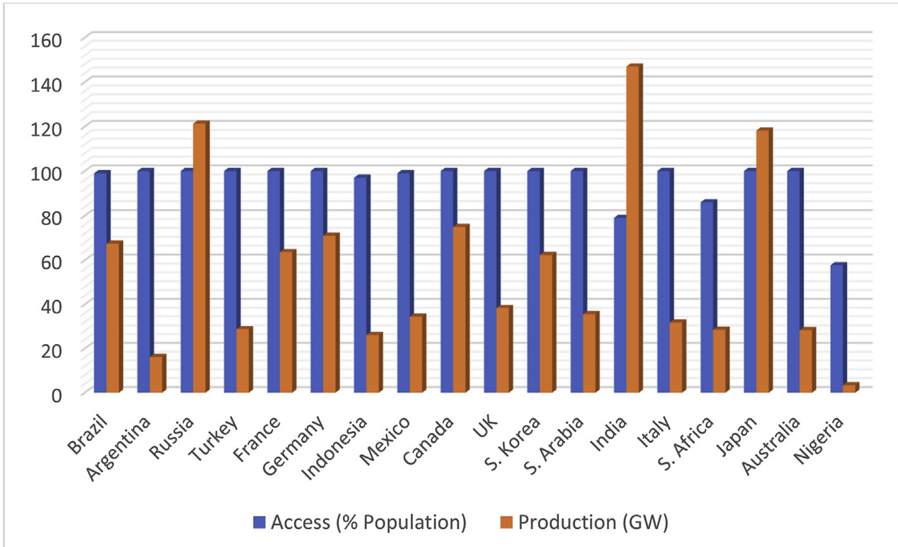

Fig. 1. G20 countries and Nigeria – electricity production and access level (Source: Author based on World Bank Data – World Development Indicators: Electricity Production, Sources and Access, 2014)

Nigeria ranks among the top countries with high solar energy potential. This potential comes with average insolation that ranges between 3.5 and 7.0 kWh/m^2/day (Okoye and Baker, 2016; INTEC, 2015; Olatomiwa et al., 2015; Daramola, 2014). Further, the Nigerian government has shown interest in renewable energy penetration to undermine the national power challenges. Previously identified challenges to renewable energy (RE) policy implementation in Nigeria include weak government motivation, lack of economic incentives, and multiple taxation (Ajayi and Ajayi, 2013). The last two identified hindrances inhibit investments from both local and foreign investors.

One commendable commitment shown by the Nigerian government is the establishment of RE feed-in-tariff regulation, which was signed into law late December, 2015 (INTEC, 2015). This law can be viewed as a case for breaking barriers to power generation through full use of available RE resources. However, before analyzing policy support for RE, this article first investigates the viability of RE technology adoption by private businesses in the country.

2. The electricity sector in Nigeria

In 2016, per capita power consumption in Nigeria was reported to be one of the lowest globally, at just 151 kWh. However, the country has the potential to reach 982 kWh per year by 2025, especially through accelerating growth in generation capacity (PWC, 2016). The Nigerian power sector has undergone various transformations beginning in 2005 with the establishment of National Electricity Regulation Commission (NERC) and the formation of the Power Holding Company of Nigeria (PHCN) alongside the new electric power reform act. Fig. 2 shows the transformation process between 2005 and 2015. The initial reforms aimed to address the inefficiency of the incumbent power authority, the Nigeria Electric Power Authority (NEPA). This led to the unbundling of the sector, giving birth to 18 companies, 6 generation, 1 transmission and 11 distribution companies as successors. This transformation period also involved strengthening the renewable energy program in 2014, with aggressive plans to harness solar energy resources. However, unbundling of the sector over the past 10 years has not culminated into the expected improved electricity supply (PWC, 2016).

Over the next decade, Nigeria must look towards improving capacity utilization (currently at 31%) significantly by investing in new and more efficient power generation technology, as well as revamping existing power plants (PWC, 2016). More so, there is high imbalance between generation capacity and transmission capacity of up to 8.6 GW deficit. The Nigerian electricity generation mix is comprised of twenty gas powered generation facilities and 3 hydro power plants with total installed capacities of approximately 8.4 GW and 3.65 GW respectively (ICREEE, 2016). However, many of these power plants are non-operational. As of February 2016, the peak generation from all power plants was 5.07 GW (Akwagyiram and George, 2016). Meanwhile, a demand of over 24 GW was estimated for this same period (INTEC, 2015).

Nigerian electricity generation can be classified into four groups based on grid connection or off grid, consumption, and permit and license requirements (INTEC, 2015); they are captive, off-grid, embedded, and on grid generation. This study focuses on captive generation by Nigerian private industries. The main source of captive power generation is refined petroleum products like automotive gas oil (AGO) also known as diesel, premium motor spirit (PMS), and a very few with solar inverters. Captive generation with DG now serves as the main supply of electricity. Off-grid generation is usually deployed where grid connections are unavailable or not accessible such as in rural communities of the country (INTEC, 2015). Installed capacity ranges from 0 to 1 MW (Bagu et al., 2016). Users are mostly village households, governments, and private facilities within the locality (INTEC, 2015). Ninety-three million Nigerians, over 50% of the population, are reported to be off the grid (INTEC, 2015) (Anyagafu, 2014). Embedded generation involves a grid-connected system using distribution belonging to an external distribution company. NERC and power purchase agreement (PPA) are requirements to operate and electricity generation below a certain threshold cannot be transmitted on high voltage lines. Distribution lines in Nigeria are of 11 kV and 33 kV. Currently, grid connected electricity generation is composed primarily of petroleum products, 39.8% gas, 24.8% oil, 0.4% coal and then about 36% from hydro (Aliyu et al., 2013). The energy value chain includes the limitations of transmission and grid access rates. Decentralized systems in the form of captive power from petroleum products is the current order of the day in the country.

3. Methods and data analysis

To investigate the suitability and viability of hybrid diesel and solar PV systems for the Nigerian private sector, case research method was used. Standardized survey questionnaire was designed to collect both qualitative and quantitative data. Private companies in Nigeria were invited to complete the survey via telephone, physical, and e-mail contact. The survey process involved sending the questionnaires to

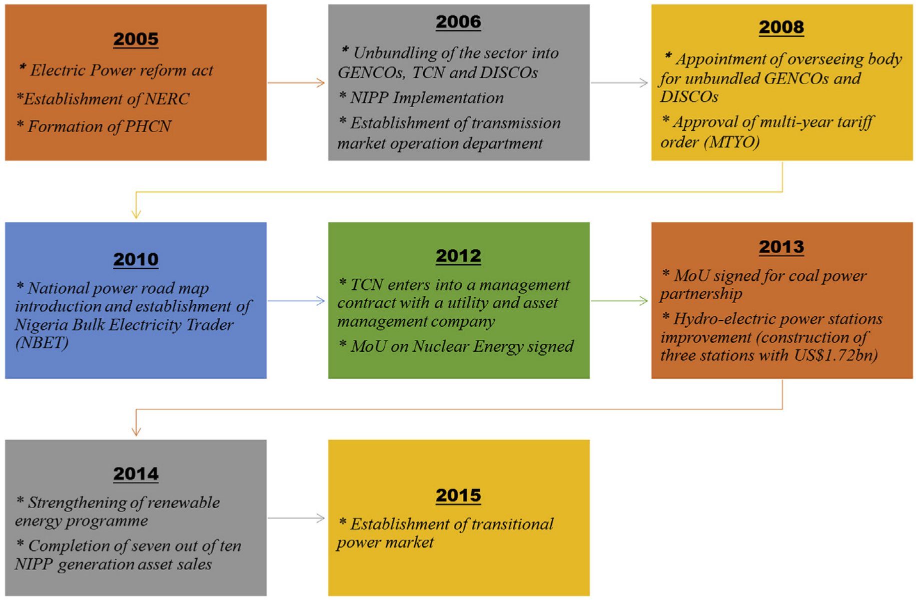

Fig. 2. Nigerian power sector transformation (Source: Authors based on PWC, 2016 report).

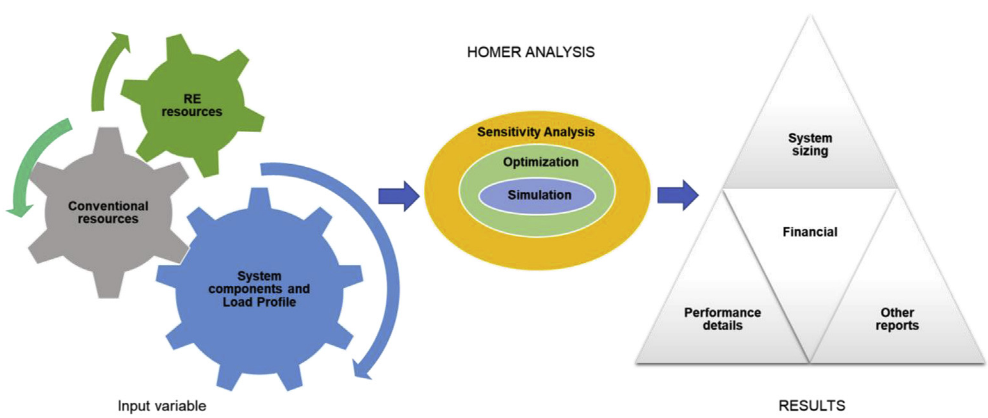

Fig. 3. Schematic description of HOMER modelling and simulation processes (Source: Authors).

about 150 companies spread across Nigeria, out of which 40 responses were recovered. Although, this represents a response rate of approximately 27% with non-respondent bias of 73% (Fincham, 2008), it also means a margin of error outside acceptable returned sample size for a continuous data (Kotrilik and Higgins, 2001). However, survey responses covered businesses in telecommunication, health, manufacturing, energy, real estate, hospitality, education, banking, and service. Seventy (70) percent of the respondents represent the health care, banking and downstream energy sector of the country. Data extracted from the survey were used as inputs for model design and simulation in HOMER®. The distribution of respondents is such that approximately 87% of the 40 responses are from Lagos state, 5% from Ogun state, and 3% from each of Abuja, Imo and Rivers state. Only 5 out of 36 Nigerian states were covered in the responses received. This can be explained by the peculiarity of Lagos being the most industrialized and commercial city in Nigeria, and the research may thus be especially suited for companies in Lagos.

Electricity tariff data of NERC on Nigerian current multi-year tariff order (MYTO) (for those connected to the grid supply) was used to shape the system simulation for LCOE comparison. The reason is that electricity tariff varies with distribution company (DISCO) operating in a region in Nigeria. The quantitative data collected in this study included features such as load profile, energy and fuel consumption, cost and sources of electricity. Data included the name and location of businesses, as HOMER uses the location name to generate geographical coordinate system to obtain corresponding resources such as global horizontal irradiance (GHI) and temperature. Space available for PV installation was assessed, as was connection status; grid supply reliability; fuel type and annual consumption; number, capacity, and age of existing generator systems; generator operations; costs associated with operation and maintenance of generator systems; hourly load profiles; and monthly energy expenses for grid connection customers. The survey also collected data from respondents regarding knowledge of RE, perceived effectiveness of present energy system, effects of changes in fuel price and willingness of the organization to invest in energy technologies like solar PV.

HOMER Pro 3.9 is the global standard in microgrid energy simulation software. HOMER® was mainly used for systems modelling and obtaining simulation results. The software is designed with capacity to model a combination of traditionally generated and renewable energy power systems, including storage systems and load management. It considers systems complexities and builds the most cost-effective solutions in optimization with predefined input variables. The software also has capacity to simulate thousands of comparable systems depending how it is setup. Fig. 3 shows a representation of the modelling process.

4. Results: social, technical and policy feasibility

This section presents qualitative analysis of survey and simulation results examining the sociotechnical and policy feasibility for PV diesel hybrid adoption among private industrial actors in Nigeria. Results obtained reveal that current energy cost (LCOE) with DG is less economically efficient for Nigerian businesses compared to the cost when PV is added. Furthermore, it shows high willingness of private businesses to invest in more cost effective and secure energy systems from renewable energy sources. Review of existing energy policies in Nigeria reveals the country's intention to support development of the renewable energy generation sector. However, these policies are arguably inadequate to take the country's energy sector to the next level. Thus, a review of possible policy options for maximizing renewable energy integration in the Nigerian private sector is provided.

4.1. Social feasibility

This research examined the current concerns of private business energy consumers by interrogating their level of dependence of DG, the impact of this on their business, and their willingness to switch their energy system. Only eleven of the companies surveyed are connected to the grid for supply of electricity either as main source or standby, meaning that 72% of these responding companies depend solely on captive diesel generators for their daily business activities. When asked about satisfaction in their current energy situation, over 80% are unsatisfied, reporting high operation costs as well as regular changes in price of diesel as reasons for their dissatisfaction. Companies that are connected to the grid or off-grid all face these challenges, as there is very close similarity in daily operation hours of diesel gensets in each category.

The research also attempted examine levels of knowledge regarding renewable energy considering the current low deployment across the country. The surveyed companies were asked their level of agreement with the claim that there is limited in-house knowledge of RE (including solar PV), and 21 respondents out 40 agreed with this hypothesis, while 15% could not ascertain how much they know about the technology, as shown in Fig. 4. By implication, this shows that few private businesses are aware of the potentials of harnessing RE resources for power generation.

Furthermore, 25 respondents (63% of total) say they are willing to invest in alternative energy sources that are more cost effective and 31 respondents (approximately 78% of the respondents) declared that the shortfall of power supply has adversely affected their business. From these responses, it can be argued that the lack of adequate information on RE potentials is limiting the ability of these companies to invest in better power supply sources other than DG.

The survey on monthly fuel expenses also reveals that as much as €4.2million goes into annual genset fueling expenses for 39 out of the 40 companies. Using Central Bank of Nigeria exchange rate of NGN 341/€ for 20th June 2017 (CBN, 2017), this represent an average of €107,610 per annum. A greater percentage of this cost, up to 87% comes from 25% of surveyed companies, and they are all not connected to the grid. This is excluding annual maintenance costs of these DGs, with an average of €6,000 from 32 of the respondents provided this cost. Only 33% of respondents indicated that they are aware of RE as a possible solution. However, 63% are willing to invest on alternative energy solutions such as solar PV if they provide better solutions (also in Fig. 6).

4.2. Technical feasibility

To make an argument for RE utility for Nigerian private businesses, it is expedient to investigate technical feasibility. Three different cases were considered in assessing the viability of solar PV-DG hybrid systems for energy supply among Nigerian private businesses. Simulation results show huge savings with the hybrid system relative to the current scenario, otherwise defined as business as usual (BAU). In addition to solving environmental problems of CO_2 emissions, the hybrid system also offers possible financial incentives for companies that can trade their emission. The value of revenue from emission reduction is also computed in this chapter.

From a total of 40 surveys, only 28 companies provided all required data needed for modelling. These 28 respondents were clustered into three categories to represent systems with different monthly load factors: low, medium and high load factors as shown in Fig. 5. The lines drawn on the graph indicates the boundary between one cluster and another. The load factor for each system was computed using the formula relationship between average consumption, peak demand, and billing cycle values (Power Planet n.d).

$$\text{Load Factor, } L_f = \frac{\text{Average Monthly Consumption (kWh)}}{\text{Peak Demand (kW) x 30days x 24hr}}$$

Average Monthly Consumption = (average load/day) x (working hours) x (days/month)

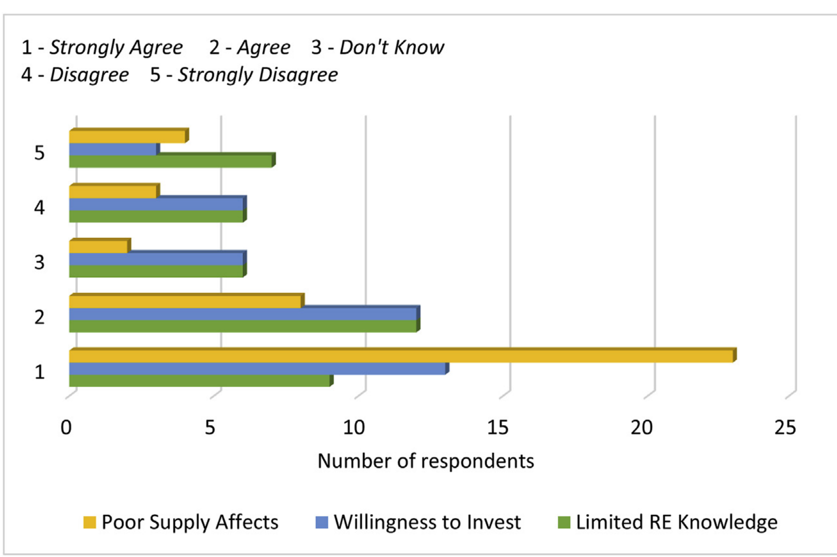

Fig. 4. Survey responses.

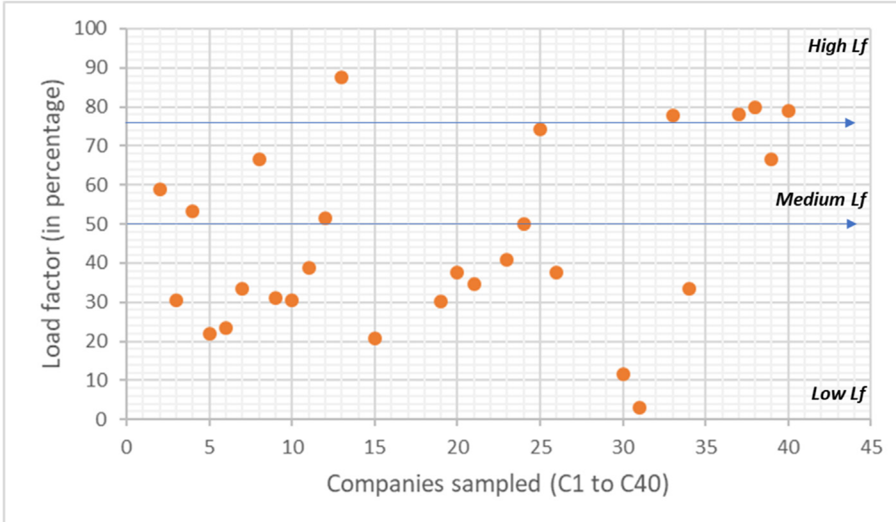

Fig. 5. Load factor clustering.

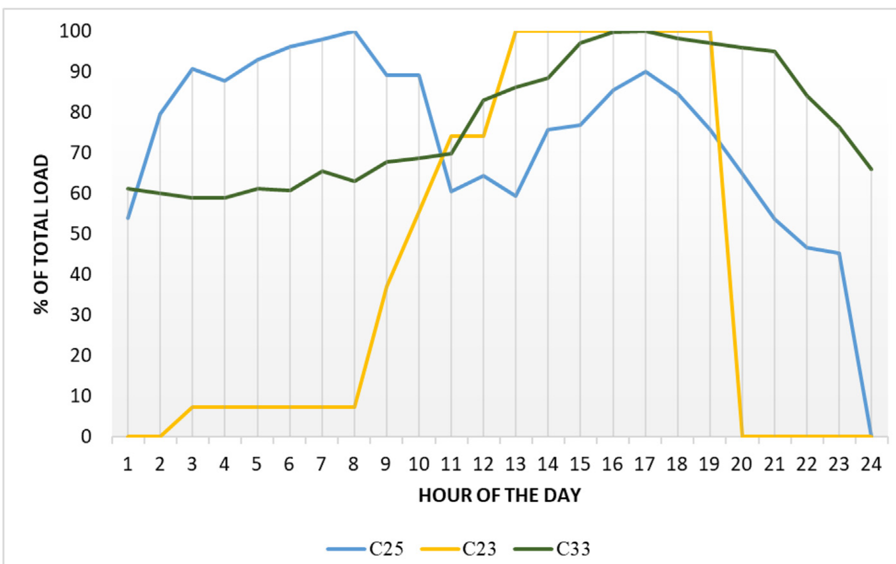

Fig. 6. Electricity usage pattern of the three case study.

From these clusters three companies, C23, C25 and C33, were chosen at random from each group. The three chosen companies provided full load data as well as other information that can be modelled in HOMER. Fig. 6 shows electricity usage patterns, being percentage of total (peak) load of each company. Their load sizes also vary increasingly in that order. The low load factor category, having value less than 50% (or 0.5), includes 16 out of 28 companies, based on data provided. One from these 16 private companies, labelled as C23, was selected randomly and modelled. The daily consumption pattern from the hourly load profile (Fig. 6) shows that the system is operated on a consistent daily schedule, as the load is constant between 12hr to 18hr and then drops to zero by 20hr.

C23 electricity supply is from both the grid and from DG, with peak load of 27 kW and average demand of 266 kWh/day. Medium load factor systems are defined by companies whose load factor is greater than or equal to 50% but less than 75%. This category includes 6 out of 28 companies, from which C25 with a load factor of 0.74 was selected and modelled. The company operates on an almost evenly distributed consumption in each hour throughout a 24 h period. The system is not connected to the grid and it runs wholly on DG power generation, with a peak load of 261 kW and average demand of about 4598 kWh/day. High load factor (75% and above) systems characterize fives of the respondent companies. One of these, C33 has load factor of 78% and was randomly selected. This system has a peak load of approximately 4.6 MW and average demand of 85.3 MWh/day. The organization's energy system represents a very high load for supplying a big community having many other institutions, small and medium-scale enterprises (SMEs), and residential buildings. C33 also runs on 100% DG power supply.

To assess these three cases, two scenarios were developed. The BAU scenario considers the present energy status with respect to system configuration, electricity demand, cost, GHG emissions for all the systems in previously selected. The PV-DG hybrid scenario considered the addition of solar PV into the system. Comparison of results of these two scenarios reveals conditions in which the hybrid system is most suitable. The BAU input parameters used in modelling and simulation are data provided in the survey which include annual generator O&M cost, connected grid or off grid system, fuel type and delivery prices, grid prices (for grid connected system), grid reliability (power outages), and usual daily operating hours of DG system. Also, the hourly load profile and the number of AC generators used by each company was included in the model.

However, certain assumptions were made to complete the modelling and carry out system simulation. Firstly, the capital cost of DG systems in the model for all the cases was taken to be zero. This assumes that current generating sets of these companies can operate at needed capacity to meet demand. Replacement cost of the generator was set at €300/kW based on online market prices (Okafor, 2016), with price comparable for varying capacity (kVA). Generator market prices were converted to euro (€) equivalent. Furthermore, the modelling was carried out using an auto-size generator for systems with single genset and a combination of this with another matching capacity genset in the catalogue for systems with more than one unit.

An auto-size genset is a generic plant capacity designed by HOMER to match the prescribed load size of a system. For DG O&M cost, this constitutes the integration of material, logistics, spare parts, labor and other associated costs that ensure that the unit keeps running (Oladokun and Asemota, 2015). Other considered inputs are load variability and grid price of 0.067€/kWh (equivalent to NGN 22.96/kWh) from MYTO of Eko Electricity Distribution Company (EKEDC) was used for Case I. Annual capacity shortage was also assumed to be 0% and inflation rate and nominal discount rate were set at 12% and 14% respectively, generating a real discount rate of 1.79% (Trading Economics, 2017). A project lifetime of 25 years was assumed. More so, each system is taken to be operating optimally and as such optimal simulation results from HOMER® were assumed.

For the PV-DG scenario, data for simulation and system optimization include the size of direct current (DC) connected generic flat plate PV component, solar GHI and temperature as resources (which is NASA 22-year data), system converter for PV, and converter component cost. The cost of PV forms the core of the project's viability assessment; thus, much care was taken to ensure that a close to real cost was assumed for the three case studies under consideration. Based on project cost in West Africa as reported in an IRENA report (IRENA, 2016), €1,900/kW was assumed for each of capital and replacement cost of PV. Converter capital and replacement cost of 10% PV cost was assumed, which equals €190/kW each. O&M cost of €10/kW and €0/kW was assumed for PV and converter respectively. The model design also assumes a project lifetime for solar PV of 25 years and 15 years for the inverter input with 95% efficiency.

4.2.1. Simulation results

Simulation results of BAU for the three cases are presented in Table 1, showing the current system architecture, fuel price, O&M cost, grid price (for system connected to grid and receiving billings), annual consumption, LCOE and CO_2 emission. The LCOE of C23 is the lowest among the three cases. One reason for this is that this company is connected to the grid, which provides a very cheap tariff. With the current Nigerian MYTO, energy costs for commercial and industrial setups on the grid appears to be more competitive and attractive. However, low reliability and availability of supply from the grid, which is a major setback for private businesses, requires that a better alternative energy solution be explored. C33 modelling results in a LCOE of 16 cents per kilowatt compared to C25 with approximately 17 cents per kilowatt. One inference that can be made from this BAU scenario modelling is that companies operating with a higher load factor have lower LCOE than others, except for grid connected system in the mix. GHG emission, especially CO_2, is proportional to the size of energy consumption from DG.

From the simulation results of hybrid scenario (Table 2), annual energy production for Case I is 99,955 kWh: 29% from solar PV, 37.1% from genset and the remaining from the grid, with 1.47% excess electricity. Annual generation for Case II includes 83% penetration from the two gensets and about 19% from solar PV, for a total of 1,756,380 kWh. Excess generation of 3.8% was observed in the system. Case III modelling results in an annual production of approximately 32 TWh with 76.9% penetration by DG and 23.1% from solar PV. From consumption, an excess of 707 MWh/year or 2.2% of the total generation is predicted.

Savings from solar PV introduction into the hybrid energy system can serve as a source of revenue for Nigeria. For the three case studies, the savings that can be achieved is calculated as the difference between LCOE of PV-DG hybrid system and that of Business-As-Usual scenarios as shown in Table 3. The savings are €491, €3,357 and €280,355 per annum for C23, C25 and C33 respectively.

4.2.2. Sensitivity analysis

To arrive at a real system cost and economic comparison in each scenario, certain factors that are susceptible to changes throughout system lifetime need to be checked irrespective of the energy source. Sensitivity analysis applied to the models in this research consider three significant varying factors such as annual increase in O&M cost, diesel price volatility and capital cost of genset with respect to system LCOE. Other factors are kept constant to reduce model complication.

4.2.2.1. Annual increase in O&M cost. From the survey conducted about 50% of the respondents could provide annual genset O&M cost and an estimated annual increase in this cost. Other respondents had the space for this question unanswered. Responses given show a very wide range of increment rate for this cost, ranging between 10% to 130%. However, genset O&M cost annual increment has been found to increase geometrically with capital cost of a unit in which 8.8% of the capital cost is taken to be the O&M cost and can be used for the first year of the genset (Oladokun and Asemota, 2015).

Subsequent years are computed using a more generally representative formula (Oladokun and Asemota, 2015) was used to estimated annual €/hr O&M cost as follows;

$$OMC_n = \beta CAT(1 + g)^{n-1} \text{where } n = 1; 2 \ldots N$$

OMCn is the cost of operation and maintenance for n year
CAT represents the capital or acquisition cost of the genset
βis defined as O&M factor which is taken to be 0.2
g is 8.8% defined as percentage of the capital cost of genset
N is the last year of operation of the genset

Thus, using the capital cost of €300/kW, Fig. 7 shows estimated annual O&M cost in €/hr of a DG unit for its first 25years.

Matching the age of genset indicated by each company in the survey Table 4 shows result of computed O&M increase in €/hr, in the next 10

Table 1
BAU simulation result.

	Case I: C23	Case II: C25	Case III: C33
1. System Current Architecture	Genset (31 kW), Grid 999,999 kW	CAT C9 ATAAC 200ekW (200 kW), Autosize Genset (290 kW)	Autosize Genset (5,100 kW)
2. O&M Cost (€/kWh)	0.015	0.015	0.010
3. Fuel Price (€/ltr)	0.6	0.47	0.44
4. Grid Price (€/kWh)	0.067	NA	NA
5. Annual Consumption (kWh)	70,090	1,678,270	31,150,560
6. LCOE (€/kWh)	0.156	0.167	0.16
7. CO_2 emission (kg/yr)	70,998	1,269,199	21,994,913

Table 2
Hybrid result.

	Case I: C23	Case II: C25	Case III: C33
1. System Current Architecture	Genset (31 kW), Grid 999,999 kW, Generic flat plate PV (20 kW), System converter (15.1 kW)	CAT C9 ATAAC 200ekW (200 kW), Autosize Genset (290 kW), Generic flat plat PV (200 kW), System converter (102 kW)	Autosize Genset (5,100 kW), ABB with PVS980 2000 Generic PV (3,500 kW), system converter ABB PVS980 (2000 kW)
2. O&M Cost (€/kWh)	0.015	0.015	0.010
3. Fuel Price (€/ltr)	0.6	0.47	0.44
4. Grid Price (€/kWh)	0.067	NA	NA
5. Annual Consumption (kWh)	97,090	1,678,270	31,150,560
6. LCOE ((€/kWh)	0.149	0.165	0.151
7. CO_2 emission (kg/yr)	53,573	1,125,593	17,775,944

Table 3
Savings potential.

	Case I, C23	Case II, C25	Case III, C33	Total
LCOE difference between PV Hybrid and BAU (€/kWh)	0.007	0.002	0.009	
Annual electricity consumption (kWh)	70,090	1,678,270	31,150,560	
Savings (€/Yr)	**491**	**3,357**	**280,355**	**284,203**

years for case study; C23, C25 and C33. Thus, for instance in 2017 hourly O&M cost used in model simulation for the three cases respectively are €0.015, €0.015 and €0.010. Other corresponding values were used for varying years till 2017.

4.2.2.2. Fuel price volatility. Like the O&M cost, there was variation in per litre price of diesel fuel reported by all the respondent of the survey. Possible reasons for this price difference include company supplier, proximity of site to petroleum depots and market deregulation effects. Furthermore, as the country continually import a large percentage of petroleum products consumed, prices have been highly subjective to exchange rate. For instance, in 2002 the average price of diesel was around NGN 25 per litre while 2016 the average price stood at approximately NGN 190/litre. (EOSTRA, 2015). These reasons result in instability of diesel fuel price in Nigeria. A linear forecast for the next 10years was made using price trend of previous years. The result of the forecast is shown in Fig. 8

Using the assumed exchange rate, a diesel price of €1.32/litre (equivalent to NGN 450/litre) has been forecasted for the year 2027. For other years before 2027 an annual addition of €0.03 to various site fuel prices in 2017 was used in the model and simulation to check LCOE.

4.2.2.3. Capital cost addition. Inclusion of capital cost which was initially set at €0/kW in the model would produce new simulation result, mostly on the economic part. Considering that C23 and C25 DG units are as old as 10 years of daily operational use (not less than 30,000 h for each, assuming 250 working days in a year), it was necessary to consider capital cost inclusion. The high operational hours of these gensets would have reduced their capacity factor. Therefore, the model sensitivity is tested with addition of capital cost of €300/kW alongside replacement cost.

Table 4
Computed hourly O&M cost of case study for the next 10years (source: Authors).

Next 10years	Case I: C23	CaseII: C25	Case III: C33
2017	0.015	0.015	0.010
2018	0.016	0.016	0.010
2019	0.017	0.017	0.011
2020	0.019	0.019	0.012
2021	0.021	0.021	0.013
2022	0.022	0.022	0.015
2023	0.024	0.024	0.016
2024	0.026	0.026	0.017
2025	0.029	0.029	0.019
2026	0.031	0.031	0.021
2027	0.034	0.034	0.022

4.2.3. Sensitivity results

With different fuel prices and O&M costs of DG incorporated into the model, the following are simulation results of LCOE plotted against annual fuel prices (starting from 2017) obtained from optimum

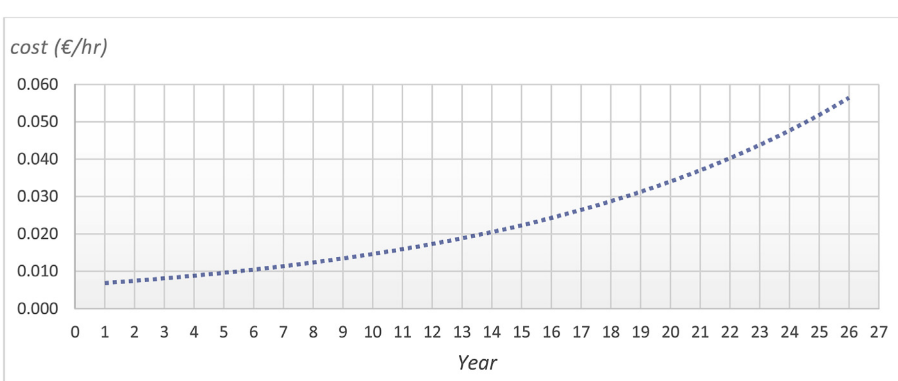

Fig. 7. Computed hourly O&M cost of DG from year 1–25 (Source: Authors).

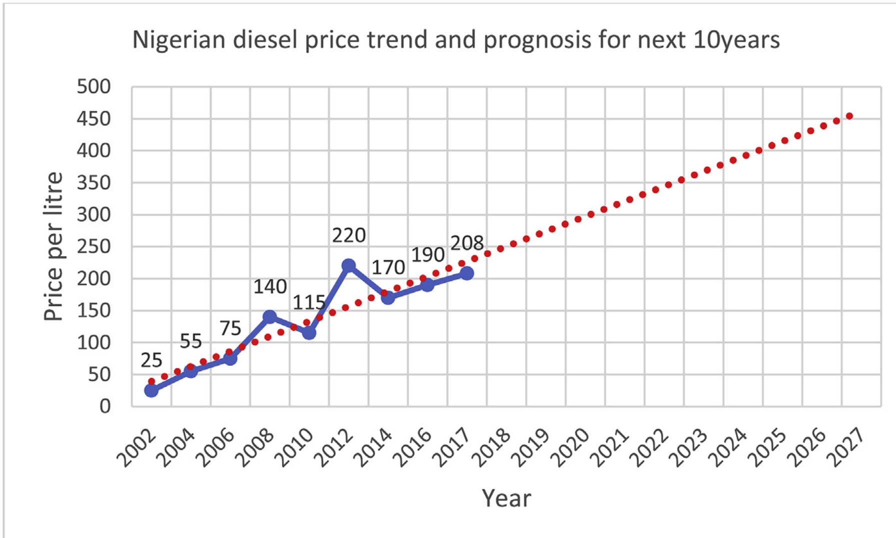

Fig. 8. Diesel price forecast till 2027 (Source: Author Based on EOSTRA data, 2015).

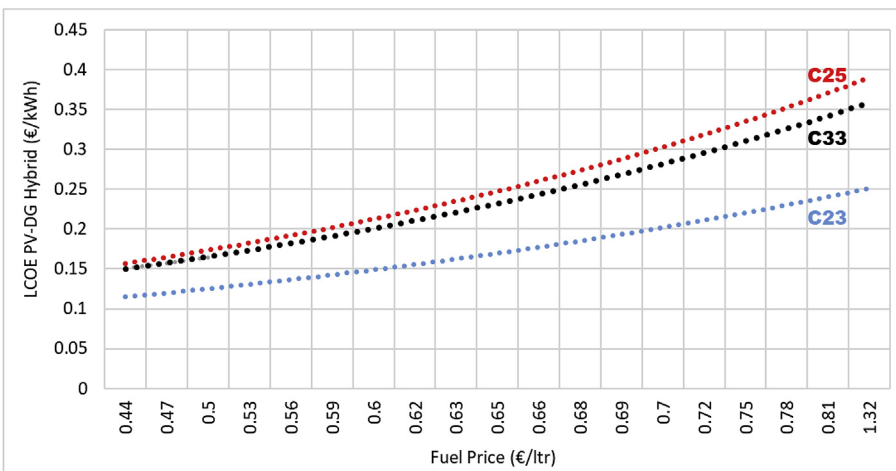

Fig. 9. Sensitivity analysis summary for the three companies (Source: Author).

solutions for each case. Firstly, results for all the cases (Fig. 9) show that LCOE will increase linearly with increase in O&M cost and fuel price. Grid connected hybrid system as in the case of C23, is the most competitive in LCOE compared to off-grid hybrid system. At fuel price of €0.69/L, the LCOE for C23, C25 and C33 are 0.2, 0.3 and 0.28 €/kWh respectively.

4.3. Policy feasibility

This section reviews different renewable energy policies across the globe in juxtaposition with the existing ones in Nigeria. In many developing countries, the main hindrances to RE investment and technological growth include lack of capital and adequate policy (Ata, 2015). Beyond the economic and market outlook of RE technologies, the likelihood of large-scale increase in investment or deployment is stalled in the absence of policies that reduce risk and provide access to investment capital. These policies may require tax credits, feed-in tariffs (FIT) or direct subsidies (Ata, 2015).

The subsidies being referred to here is investment subsidy rather than electricity subsidies (Ikeme, J. and Ebohon, O. J., 2005). FIT is a regulatory instrument that provides investment guarantee to power plant developers through a buy back price of power fed directly into the grid (World Energy Council, 2016). It is usually designed for a fixed period to ensure regulated return on investments. Another key policy that would be vital for developing countries to adopt is the concept of renewable portfolio standard (RPS). RPS has played quite a big role in the development of renewable energy in some states in the United States (Schelly et al., 2017). Policies such as RPS at state level would serve as clean development mechanism (CDM) promotion (Thiam, 2012).

In the last few years, the Nigerian government has shown a growing interest in RE development through different policies, programs, and partnerships. These policies include National Electric Power Policy (NEPP), National Energy Policy (NEP), National Power Sector Reform Act of 2005, Renewable Energy Master Plan (REMP) in 2005 and National Renewable Energy and Energy Efficiency Policy (NREEEP) in 2014 among others (Nnaemeka and Ebele, 2016). The REMP incorporates renewable energy feed in tariff policies which was signed and sealed in 2015. Nigeria also ranks among the first nations globally to adopt SE4ALL, with the launch of the initiative since 2012. The SE4ALL initiative's main objectives include increasing RE share in global energy mix by 2030 (see Fig. 10), improving energy efficiency, and promoting universal access to modern energy services (ICREEE, 2016). Key determinants to the attainment of SE4ALL goals are resource availability, current energy supply and demand level, policies and regulatory regimes. With these pre-requisites abundantly available,

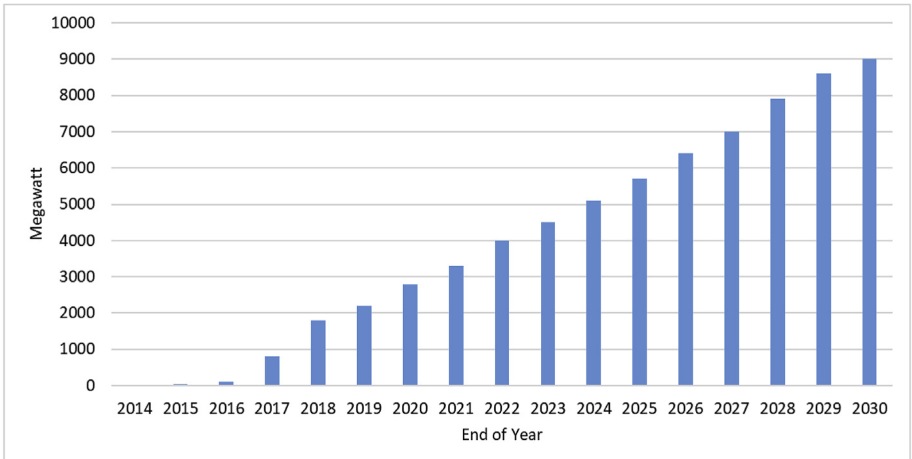

Fig. 10. Nigeria RE Targets (Source: Author based on ICREEE, 2016, p.41).

Nigeria is arguably capable of achieving all the SE4ALL objectives through proactive implementation (NERC, 2015a,b; Anyagafu, 2014).

The objectives of SE4ALL overlaps with the NREEE policy design for the Federal Republic of Nigeria. The policy framework was put together in collaboration with many Nigerian ministries, stakeholders, and international partners. The country's RE target is to ensure a share of 30% of RE in the electricity mix by 2030; 47% from hydro (both small and large) resource and 39% from solar energy source (NERC, 2015a,b; Anyagafu, 2014). Fig. 10 shows the gradual process designed to achieve the vision. Each capacity target in MW is to the end of year (EOY) deliverable. To further encourage private investors in the development of the sector, up to seven (7) years tax holiday can be available to private energy companies including utility services (both conventional and RE), electrical component manufacturers as well as solar energy powered equipment producers (ICREEE, 2016).

To further emphasize its commitment, the Nigerian government through its energy commission, NERC signed into law the RE feed-in-tariff (referred to as REFIT) in 2015 (NERC, 2015a,b). The action was one of the progressive steps that was made by the government on calls for enabling energy policies. The overall goal behind the FIT policy of NERC is to reduce energy poverty and improve utilization of abundant energy resources laden in the country. The objective of this law according to the commission include power supply revamping; stimulation of private sector involvement in deploying RE technologies for power generation, ensure that RE is fully harnessed for national energy mix and to attract private sector investments in the energy market as well (NERC, 2015a,b).

Solar technology was assigned the highest REFIT for the 2016 base year. Table 5 showing the rate at euro equivalent at exchange rate of €1 to NGN 341. There is a target of 1 GW grid connected RE by year 2018 (NERC, 2015a,b). Thirty-eight percent (38%) of this target is designated for solar PV technology including from hybrid systems.

Fig. 11 shows a summary of qualification requirements according to the REFIT regulation as published by NERC. This REFIT is only applicable for grid connected power plants. The minimum capacity requirement to be qualify for the REFIT is also pegged at 1 MW. According to the regulation as signed by NERC, the cost component for REFIT depends on technology type, estimated lifetime and amount generated by power plant. It considers costs such as plant investment cost; development, construction and connection costs; all O&M costs, fuel cost (applicable to biomass technology only) and financing cost commensurable to invested capital.

Table 5
REFIT rates in Nigeria (Source: Authors based on NERC, 2015a,b data).

Description	Solar	Wind	SHP	Biomass
Capital cost (€/MWh)	103.72	72.70	90.58	65.69
O&M cost (€/MWh)	0.09	0.89	0.16	25.05
Total (€/MWh)	**103.80**	**73.59**	**90.74**	**90.74**

The REFIT cost for solar technology for 2016 base year was US$ 177/MWh (at exchange rate of NGN 200/US$), covering capital cost of US$ 176.85 and the rest as O&M cost. The equivalent of this tariff in euro is €103/MWh, subject to review every three years with resulting effects on new plants. Interestingly, hybrid systems with any of the stipulated RE components that is connected to grid and meet the capacity requirement have opportunity to benefit from the tariff. Thus, private businesses having solar PV and DG hybrid energy system are potentially suited for this REFIT. More so, all the 11 DISCOs have been allocated predefined MW of grid fed electricity as take-off from RE power plants according to NERC policy. Thus, there is guarantee of investment returns through power sales of generation that is fed into the grid.

Net metering is also another incentive for investors in RE technologies in NERC's policy. The concept of net metering is to allow excess energy from captive generation, specifically from renewable sources to be sold to the grid at the stipulated grid price. In Nigeria, capacities below the stipulated minimum for REFIT are subjected to net metering conditions and renewable component of hybrid installation that meet the REFIT requirement are acceptable as well. Thus, grid connected RE producers primarily for self-consumption would have less to worry on their excess generation as it can be a source of revenue. However, the Nigerian populace seemed to be unaware of this policy and its accompanying benefits.

4.4. Environmental implications

The environmental impacts of conventional electricity generation (excluding nuclear power) are well known, including climate change and health impacts (Roche et al., 2017) resulting from continuous emission of greenhouse gasses, which are dangerous to humans and the entire environment. Increasing average temperatures of the earth has also attracted global concern. The COP21 consensus adopted at the 2015 UNFCCC on limiting global warming below 2 °C is one major risk to continuous use of fossil fuel (World Energy Council, 2016), which Nigeria needs to consider in making future energy plans.

Poisonous carbon monoxide released from exhaust of diesel gensets pollutes the air and found in large quantity in Nigeria, threatening lives of its citizenry (Roychowdhury et al., 2016). Emission gases include CO_2, CO, SO_x, N_2O, and methane. CO_2 release into the atmosphere is a major source of global warming, leading to climate change. Diesel generator's emission (g/kWh) of CO_2 is considered the highest among

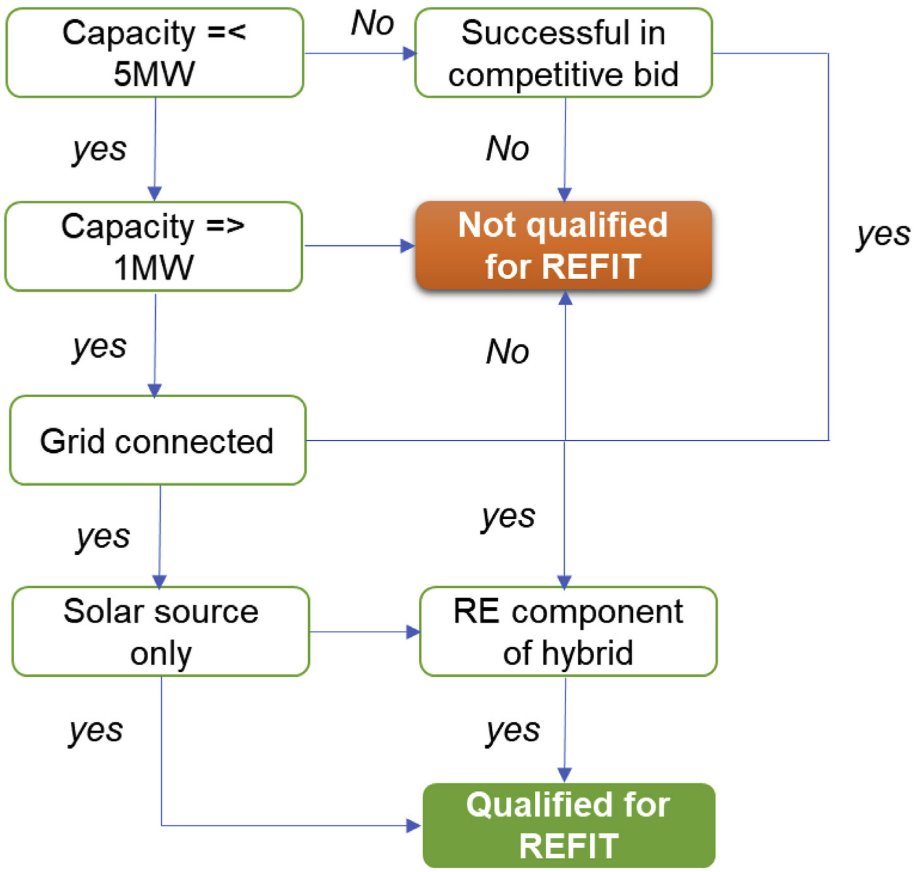

Fig. 11. REFIT requirement and qualification (Authors based on NERC, 2015a,b FIT document).

Table 6
Plants emission (Moss and Gleave, 2014)

CO_2 Emissions	Quantity (g/kWh)
DG – small size (< 60 kW)	1580
DG – medium (> 60 kW and < 300 kW)	883
DG – large (> 300 kW)	699
Other Power Plants	
Coal	969
Oil	792
Natural gas	553

other power plants like coal, oil and natural gas, especially from small plants. Table 6 shows different power plants and their CO_2 emission quantity. These small and medium size plants are used by private SMEs in Nigeria for power supply.

Based on CO_2 emissions released, Nigeria has been ranked as 46th in the world (Adebayo, 2014). Nigerian CO_2 emission figure in 2013 was 95,650 kt, representing 12.2% of total from Sub-Sahara Africa (The World Bank, 2013), requiring appreciable attention to carbon footprint. As the country's economy grows and become more industrialized, this emission would continue to scale up unless conscious efforts are made to curtail it. This can easily be achieved with increased use of clean energy technologies such as solar. Among all other energy resources, solar energy has the lowest level of GHG emissions (World Energy Council, 2016). One of the impacts of more penetration of clean energy such as PV-DG hybrid system from the massive private sector is reduced emission of GHGs.

4.5. Policy implications

When there is no public policy and regulatory framework for RE, in contrast to the case of the energy market in Germany (Reuter et al., 2012), investment becomes unattractive, posing high risky. This paper section presents some additional policies in line with existing ones that can facilitate private sector investment on PV. Policies suggestions are in line with the result of simulation carried out on the energy systems of the three case studies. An outlook into solar PV financing possibility through bank loan is assessed. Furthermore, market instrument for carbon trading is introduced as a possible investment facilitator. However, the success of the policies is contingent to strong economic, social and political structures that abhor corruption, social unrest, or close doors especially on funding opportunities for energy investors. These has been seen revealed to be an indisputable pre-requisite to energy prosperity in Sub-Saharan countries (Thiam, 2011).

4.5.1. Financing policy for private investor

One of the main challenges with which RE technology adoption encounters globally is the issue of project financing because of huge initial capital involved (Mo et al., 2016). To reduce the risk level of financial institutions, government intervention is critical. CBN in collaboration with various banks can reduce lending rate for private companies interested in solar technology investment through a lease to own program. In Nigeria, Asset Management Company of Nigeria (AMCON) has been very efficient in debt recovery of bank loans by business organizations including oil and gas assets. Government support can stimulate increase in solar PV investment. This support can also be in form of allotting tax from oil and gas operation into re-investment in RE for private sector through solar loan program. The ripple effect of success of private businesses is on national GDP and microeconomy. Thus, a financing policy that brings about a win-win situation must be designed.

Table 7
CO_2 Abatement revenue.

Carbon credit	Case I, C23	Case II, C25	Case III, C33	Total
CO_2 Abatement (kg/yr)	17,025	143,606	4,218,969	
CO_2 Abatement (t/yr)	17	144	4,219	**4380**
Market price ($/t$CO_2$)	10	10	10	
Carbon revenue ($/yr)	**170**	**1,440**	**42,190**	**43,800**

4.5.2. Policies on market instrument

Nigeria can take a cue from existing market-based policy instruments on emission trading scheme (ETS) of countries such as China (Mo et al., 2016; Tu et al., 2018). ETS is a market activity that allows emissions reduced though the use of low carbon technologies to be traded. As part of her contribution to carbon reduction, the Nigerian government can incorporate a policy that requires private businesses trade of their emission from electricity generation. This can serve as another revenue generation possibility for companies on solar PV-DG hybrid system is from carbon trading. CO_2 abatement in kg/yr represents the difference between BAU CO_2 emission and that of PV-DG hybrid scenarios from simulation results.

The savings from emission is calculated taking carbon market price of 10 USD/CO_2e (Heinrich Boell Stiftung, 2017; Roche et al., 2017) as applicable to non-OECD for emission reduction. Companies can leverage the opportunity to trade their emission permissions while simultaneously meeting their electricity demand effectively and efficiently. This market-based instrument can ensure reduced emission and environmental protection. The emission cost is also taken to as one of society cost of electricity (SCOE), which includes environmental pollution, its health effect and climate change (Roche et al., 2017). Table 7 is the result of potential emission savings as well as revenue that can be achieved from the case studies of this research.

The total annual CO_2 abatement with PV-DG scenario equals 19% of total emission from the BAU of the case study. Total carbon revenue of the three companies is US $43,800. Using the CBN exchange rate US $1.12 to €1 (20th June 2017) gives total carbon revenue from the three cases as €39,107 (equivalent to NGN13.3 Million) per annum. This amount can pay annual salaries of fifteen (15) graduate employees in Nigeria. C33 with biggest generation capacity has the highest revenue benefit from trading carbon. On a larger scale, it can be inferred therefore that solar PV-DG hybrid can achieve up to 18,174 ktCO_2 emission from Nigeria's total in 2013 – translating into €16.2 Million (NGN 5.5 Billion) annual revenue. Trading emission can serve as a great source of income for Nigeria, especially the private sector.

4.5.3. Awareness policy

Based on the survey results done showing private sector's high willingness to invest on alternative energy and the economic results obtained from model, more awareness creation is necessary to bring about substantial solar PV adoption in Nigeria. There is high possibility that with more information on the cost-benefit balance and financing options for interested investors, solar PV-DG hybrid system deployment would increase greatly in Nigeria. Therefore, RE energy experts, researchers, investors and solar energy business companies within the country needs more campaign to make the full cost and benefits of solar technology known to create a robust market in Nigeria. On a little more advance level, solar is being deployed by micro scaled enterprises such as a Lagos barbing salon among others (Heinrich Boell Stiftung, 2017). This was a pilot project achieved through a pay-as-you-go (PAYG) scheme by an indigenous renewable energy company in Nigeria. This scheme will offer those business owners stable electricity and better planning if they do not default in payment. PAYG is one of the innovative renewable energy financing schemes which has more popularity in East Africa (Heinrich Boell Stiftung, 2015).

5. Summary and discussions

These research results demonstrate that for the private sector in Nigeria, the current energy situation is insufficient and there is social, technical, economic, and policy feasibility to support increased proliferation of PV hybrid systems to support adequate energy access. Private companies can benefit from the economies of solar PV addition to their present energy system. Savings from PV-DG hybrid system increases as price of diesel fuel in Nigeria trends up. With the right policy framework, poor energy access should become a history in the country as there is increase interest and willingness of private businesses to deploy more efficient energy options. The Nigerian government has ample examples of policy actions as benchmarks achieve its RE target.

Commitment needs to be backed up by definite actions in a transparent manner and devoid of being a mere political statement. The REFIT has not proven to be self-sufficient in driving desired rapid RE implementation and corresponding investments. To facilitate achievement of the REMP for year 2030, individual states might need to adopt the RPS as a fraction of their obligation to the national energy goal. On the one hand, RPS and REFIT might be viewed as policy duplication as they are supposedly meant to achieve same objective. On the other hand, it is capable of driving action at grass root level where investors and interested private businesses develop projects through state governments. With the RPS, the government can provide enabling environment for both electricity producers and the other group termed “prosumers” from renewable technologies to meet up with its commitment. A prosumer is the combined captive power generator with the sales of excess to the grid. At such, many medium and large private companies might be motivated to generate their energy from technologies such as solar PV and in turn feed in excess for neighboring towns or villages. Examples of such cluster of factories and manufacturing industries are those located in suburbs like Agbara, Ota (Ogun state) and Oluyole, Ibadan (Oyo state) as well as Ilupeju industrial estate, Lagos state.

There is also the need for strong awareness programs directed towards financial institutions, energy developers, and local investors to support existing energy policies. There is still a general perception about immaturity of RE technologies which brings about pushback in funding RE projects in Nigeria based on investment risk. To bridge this gap between presumption and reality of RE investment, there must be effort synergy between the government, financial institutions, representative of Manufacturing Association of Nigeria (MAN), RE experts, and other stakeholders. Like the command and control instrument German EEG or the financial instrument in form of production tax credit as seen in USA, the Nigerian government in collaboration with private sector must show grit in fixing the national socioeconomic challenges nexus energy.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enpol.2019.05.038.

References

Adebayo, C., 2014. How Is 100% Renewable Energy Possible for Nigeria. With Assistance of Global. [Online] Available at. http://geni.org/globalenergy/research/renewable-energy-potential-of-nigeria/100-percent-renewable-energy-Nigeria.pdf Accessed 9th July 2017.

Ajayi, Oluseyi O., Ajayi, Oluwatoyin O., 2013. Nigeria's energy policy: inferences, analysis and legal ethics toward RE development. Energy Policy 60, 61–67. https://doi.org/10.1016/j.enpol.2013.05.095.

Akwagyiram, A., George, L., 2016. Nigerian Power Problems Old and New Frustrate Buhari's Economic Push. Reuters. Lagos/London. [Online] Available at. https://www.yahoo.com/news/nigerian-power-problems-old-frustrate-buharis-economy-push-140147523–finance.html Accessed.

Aliyu, A.S., Ramli, A.T., Saleh, M.A., 2013. Nigeria electricity crisis. Power generation capacity expansion and environmental ramifications.2013.09.011. Energy 61 (11 9), 354–367. https://doi.org/10.1016/j.energy.

Anyagafu, V., 2014. Nigeria Records over 10,000 Deaths through 'generator' Fumes. [Online] Available at: Nigeria records over 10,000 deaths through. http://www.vanguardngr.com/2014/08/nigeria-records-10000-deaths-generator-fumes, Accessed date: 11 September 2017 Accessed.
Asu, F., 2016. Power generation at Egbin drops by 84% to 172MW. Business and economy. [Online] Available at: Power generation at Egbin drops by 84% to 172MW.. In: Business and Economy, PUNCH, 30th November. http://punchng.com/power-generation-egbin-drops-84-172mw/, Accessed date: 30 May 2017 Accessed.
Ata, N.K., 2015. Impact of government policies in the renewable energy investment: developing a conceptual framework and qualitative analysis. Renew. Energy Polic. 4, 67–81.
Bagu, T., Dietz, T., Hanekamp, E., Aniche, P.-E., Soremekun, B.A., 2016. Captive Power in Nigeria. A Comprehensive Guide to Project Development. *RECP Africa-EU Renewable Energy Cooperation Program. Eschborn.* [Online] Available at. http://www.euei-pdf.org/sites/default/files/field_publication_file/market_study_captive_power_nigeria_0.pdf, Accessed date: 29 July 2017 Accessed.
Bopp, G., 2015. PV Hybrid Systems and Mini Grids. *Fraunhofer ISE. Praia, Kap Verde.* [Online] Available at. http://publica.fraunhofer.de/eprints/urn_nbn_de_0011-n-4261697.pdf, Accessed date: 12 June 2017 Accessed.
CBN, 2017. Data & Statistics. Exchange Rate by Currency. Central Bank of Nigeria. [Online] Available at. http://www.cbn.gov.ng/rates/ExchRateByCurrency.asp?beginrec=101&endrec=200&currencytype=, Accessed date: 20 June 2017 Accessed.
Daramola, M., 2014. Viability of grid-connected solar PV energy system in Jos, Nigeria. Elec. Power Energy Syst. 61, 64–69.
Edomah, N., 2019. Governing sustainable industrial energy use: energy transition in Nigeria's manufacturing sector. Clean. Prod. 210, 620–629.
Elliot, L., 2014. Mint condition: countries tipped as the next economic powerhouses. [Online] Available at. https://www.theguardian.com/business/2014/jan/09/mint-condition-countries-tipped-economic-powerhouses, Accessed date: 7 June 2017 Accessed.
Emodi, N.V., Boo, K., 2015. Sustainable energy development in Nigeria: current status and policy options. Renew. Sustain. Energy Rev. 51, 356–381.
Energy, H. n.d. *HOMER Pro*. [Online] Available at. http://www.homerenergy.com/HOMER_pro.html, Accessed date: 26 July 2017 Accessed.
EOSTRA, 2015. Fuel Price Trends and Forecast. EOSTRA FLEET MANAGEMENT [Online] Available at. http://www.efm.co.za/Media/Documents/pdf/Africa-Fuel-white-papers/Nigeria-fuel-White-Paper-March-2015.pdf, Accessed date: 7 December 2017 Accessed.
Fincham, J., 2008. Response rates and responsiveness for surveys, standards, and the journal. Am. J. Pharmaceut. Educ. 2, 72. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2384218/.
Gravito, L., Pita, M., Ruiz-Cabrero, Boer W., 2016. Unlocking Nigeria's potential: the path to well-being. [Online] Available at. https://www.bcg.com/en-us/publications/2016/public-sector-globalization-unlocking-nigerias-potential.aspx, Accessed date: 23 May 2017 Accessed.
Heinrich Boell Stiftung, 2015. Energy for Nigeria: pay as you go solar. [Online] Available at. https://ng.boell.org/2015/06/10/pay-you-go-solar, Accessed date: 11 July 2017 Accessed.
Heinrich Boell Stiftung, 2017. Energy for Nigeria - True Cost of Electricity. Renewables at Work - The Solar Barber. [Online] Available at. https://ng.boell.org/true-cost-electricity, Accessed date: 22 July 2017 Accessed.
ICREEE, 2016. Federal Republic of Nigeria. *Sustainable Energy for All Action Agenda (SE4ALL-AA). Naional Council on Power (NACOP)*. [Online] Available at. http://www.se4all.org/sites/default/files/NIGERIA_SE4ALL_ACTION_AGENDA_FINAL.pdf, Accessed date: 20 July 2017 Accessed.
Ikeme, J., Ebohon, O.J., 2005. Nigeria's electric power sector reform: what should form the key objectives? Energy Policy 9, 1213–1221. https://doi.org/10.1016/j.enpol.2003.11.018.
INTEC, 2015. The Nigerian Energy Sector. An Overview with a Special Emphasis on Renewable Energy, Energy Efficiency and Rural Electrification. With Assistance of Deutsche Gesellschaft Fur Internationale Zusammenarbeit (GIZ), European Union. *(EU) and the Minitry (BMZ)*. [Online] Available at. www.gopa-intec.de, Accessed date: 7 December 2016 Accessed.
IRENA, 2016. Solar PV in Africa: Costs and Markets. International Renewable Energy Agency [Online] Available at. https://www.irena.org/DocumentDownloads/Publications/IRENA_Solar_PV_Costs_Africa_2016.pdf, Accessed date: 5 May 2017 Accessed.
Kotrilik, J.W.K.J.W., Higgins, C.C.H.C.C., 2001. Organizational research: determining appropriate sample size in survey research appropriate sample size in survey research. Info. Technol. Learn. Perform. 19.
Mo, J.L., Agnolucci, P., Jiang, M.R., Fan, Y., 2016. The impact of Chinese carbon emission trading scheme (ETS) on low carbon energy (LCE) investment. Energy Policy 89, 271–281 89:271-83.. Energy Policy.
Moss, T., Gleave, M., 2014. How Can Nigeria Cut CO2 Emissions by 63%? Build More Power Plants. Center for Global Development [Online] Available at. https://www.cgdev.org/blog/how-can-nigeria-cut-co2-emissions-63-build-more-power-plants, Accessed date: 7 August 2017 Accessed.
NERC, 2015a. Notice of Commencement of MYTO 2015. Nigerian Electricity Regulatory Commission. [Online] Available at. http://www.nercng.org/index.php/media-library/public-notices/326-notice-of-commencement-of-myto-2015, Accessed date: 7 July 2017 Accessed.
NERC, 2015b. Regulation on Feed-In-Tariff for Renewable Energy Sourced Electricity in Nigeria. NERC REFIT. Nigerian Electricity Regulatory Commission, Abuja.
Nnaemeka, Vincent Emodi, Ebele, Nebedum Ekene, 2016. Policies enhancing renewable energy development and implications for Nigeria. Sci. Educ. 4 (1).
Okafor, P., 2016. Generator Price for Petrol & Diesel of Firman Honda Mikano. [Online] Available at. https://www.naijatechguide.com/2008/03/nigeria-electric-generator-prices.html, Accessed date: 10 June 2017 Accessed.
Okoye, C.O., Baker, D.K., 2016. Solar energy potential in strategically located cities in Nigeria; Review, resource assessment and PV system design. Renew. Sustain. Energy Rev. 55, 550–566.
Oladokun, V.O., Asemota, O.C., 2015. Unit Cost of Electricity in Nigeria. A Cost Model for Captive Diesel Powered Generating System, vol. 52. Elsevier, pp. 35–40. https://doi.org/10.1016/j.rser.2015.07.028. Renewable and Sustainable Energy Reviews.
Olatomiwa, L., Mekhilef, S., Huda, A.S.N., Ohunakin, O.S., 2015. Economic evaluation of hybrid energy systems for rural electrification in six geo-political zones of Nigeria. Renew. Energy 83, 435–446.
Power Planet n.d. Electrical load factor. [Online] Available at. http//:www.demandcharge.com/Wed_Pages/Articles/Electrical_Load_Factor.html, Accessed date: 2 July 2017 Accessed.
PWC, 2016. The Power Sector in Nigeria. Powering Nigeria for the Future. [Online] Available at. https://www.pwc.com/gx/en/growth-markets-centre/assets/pdf/powering-nigeria-future.pdf Accessed.
Reuter, W.H., Szolgayová, J., Fuss, S., Obersteiner, M., 2012. Renewable energy investment: policy and market impacts. Appl. Energy 97, 249–254. https://doi.org/10.1016/j.apenergy.2012.01.021.
Roche, M.Y., Ude, N., Ofoegbu, I., 2017. True cost of electricity. Comparison of cost of electricity generation in Nigeria. Heinrich Boell Stiftung Nigeria. The Nigerian Economic Group Summit. [Online] Available at. https://ng.boell.org/sites/default/files/true_cost_of_power_technical_report_final.pdf, Accessed date: 5 July 2017 Accessed.
Roychowdhury, A., Nasim, U., Chandola, P., 2016. Towards Clean Air in Nigeria. With Assistance of Federal Ministry of Environment, Nigeria. *Edited by Archana Shankar. Centre for Science and Environment.* [Online] Available at. http://www.cseindia.org/userfiles/towards-clean-air-in-nigerian-cities.pdf, Accessed date: 11 August 2017 Accessed.
Schelly, C., Louie, E.P., Pearce, J.M., 2017. Examining Interconnection and Net Metering Policy for Distributed Generation in the United States. 2017. Renewable Energy Focus, pp. 10–19. https://doi.org/10.1016/j.ref.2017.09.002.
The World Bank, 2013. CO2 emission (kt). [Online] Available at. http://data.worldbank.org/indicator/EN.ATM.CO2E.KT?view=map&year_high_desc=false, Accessed date: 12 May 2017 Accessed.
Thiam, D., 2011. An energy pricing scheme for the diffusion of decentralized renewable technology investment in developing countries. Energy Policy 39 (7), 4284–4297. https://doi.org/10.1016/j.enpol.2011.04.046.
Thiam, D., 2012. Policy instruments for a market penetration of low carbon technology in developing nations. Energy Sect. Manag. 6 (4).
Trading Economics, 2017. Nigeria Food Inflation. 1996-2017. [Online] Available at. https://tradingeconomics.com/nigeria/food-inflation, Accessed date: 12 August 2017 Accessed.
Tu, Q., Betz, R., Mo, J., Fan, Y., Liu, Y., 2018. Can carbon pricing support onshore wind power development in China? An assessment based on a large sample project dataset. J. Clean. Prod. 198, 24–36.
World Energy Council, 2016. *World Energy Resources 2016*. [Online] Available at. https://www.worldenergy.org/wp-content/uploads/2016/10/World-Energy-Resources-Full-report-2016.10.03.pdf, Accessed date: 1 August 2017 Accessed.